What Factors Affect House Prices in New York?

Richard Li, Callista Hardi, Zhenpeng Liu, Sam Martin & Xiao Ma

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This report investigates and explores the factors that influence housing prices in New York. Using a dataset from DASL, we examined how 16 property characteristics such as living area, lot size and age affect house prices. Some type-correct changes and transformations were applied as we cleaned the data, and we used three variable selection methods (Forward, Backwards, and Stepwise) and an out of sample performance (Cross Validation) to identify key predictors. Our findings show that living area, location, and certain house features like central air conditioning play major roles in determining the price. Understanding these relationship can help predict and interpret future housing market trends in New York.

house prices | NYC

Introduction. The main question guiding this research is: What factors influence house prices in New York? Housing prices are generally affected by a combination of factors such as lot size, age, location, and various house features. Understanding these relationships provides valuable insights into how the housing market operates and may also help identify how future trends could develop.

In this report, we use the housing-prices-ge19.txt dataset. After cleaning the variables and applying logarithmic transformations to correct for strong right skewness, we aim to identify which set of variables have the greatest impact on house prices using data-driven variable selection methods and out-of-sample performance evaluations. By doing so, this study contributes to a better understanding of the factors influencing New York's housing market and demonstrates how statistical models can be used to analyse complex real-world problems.

Dataset. The dataset housing-prices-ge19.txt (DASL, n.d.) contains 1,734 observations and 16 variables describing housing attributes and sale prices in New York. The target variable is Price. Key predictors considered include Lot.Size (lot area), Living.Area (house size in sq ft), Bedrooms, Bathrooms, Rooms, Age, Land.Value, and categorical features such as New.Construct, Central.Air, Fuel.Type, Heat.Type and Sewer.Type.

On reading the file we ensured column types were correct and applied safe numeric conversion for columns containing non-numeric characters. Binary columns such as New.Construct and Central.Air were converted to factors. We cleaned up the naming conventions (Wickham *et al.*, 2019; Firke, 2024) for the variables as well as removed any empty and nonsensical entries, such as lot_size = 0. Other than that we also made sure to remove unnecessary columns like the 'test' column as it was irrelevant to the research.

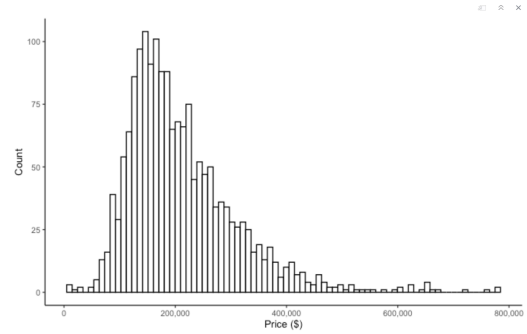

Fig. 1. Distribution house prices in New york.

Price is strongly right-skewed (median = 189.7k, mean = 211.5k, max = 775k).

Min	1st Qu	Median	Mean	3rd Qu	Max
5000	145000	189700	211545	257290	775000

Fig. 2. Statistic summary of house prices in New york.

To stabilize variance and improve linear modelling, we created ln_Price = log(Price), log(Lot.Size), and log(Land.Value). All cleaning and transformations are reproducible and included in the code appendix.

We report the univariate summary (min, 1stQ, median, mean, 3rdQ, max) for Price, show histograms for Price / ln_Price, boxplots by Bedrooms and scatterplots of Price (and ln_Price) vs Living.Area and Lot.Size. These visualizations support choice of log transform and reveal high-value outliers that will be examined in model diagnostics.

Analysis. To analyse the data, we used multiple linear regression to identify which set of variables best affects the housing prices in New York. We explored different subsets of predictors using three variable selection methods: forward selection, backwards elimination, and stepwise selection.

In model A, we used forward selection. This method starts with no predictors and variables were added one at a time based on how much they improved the model fit. The final model includes 11 predictors and has an AIC value of 685.1047.

$$\begin{aligned}\ln(\text{price}) = \beta_0 + \beta_1 \cdot \text{living_area} + \beta_2 \cdot \ln(\text{land_value}) + \\ \beta_3 \cdot \text{bathrooms} + \beta_4 \cdot \text{waterfront} + \\ \beta_5 \cdot \text{heat_type} + \beta_6 \cdot \ln(\text{lot_size}) + \beta_7 \cdot \text{age} + \\ \beta_8 \cdot \text{new_construct} + \beta_9 \cdot \text{central_air} + \\ \beta_{10} \cdot \text{pct_college} + \beta_{11} \cdot \text{rooms}\end{aligned}$$

For model B, we used backwards elimination and unlike model A, this method starts with all the predictors, and a variable with the least contribution is removed in each step. The final model also has 11 predictors, however, model B was observed to have a lower AIC of 684.4653, suggesting a better fit compared to model A.

$$\begin{aligned}\ln(\text{price}) = \beta_0 + \beta_1 \cdot \text{waterfront} + \beta_2 \cdot \text{age} + \beta_3 \cdot \text{new_construct} + \\ \beta_4 \cdot \text{central_air} + \beta_5 \cdot \text{fuel_type} + \beta_6 \cdot \text{living_area} + \\ \beta_7 \cdot \text{pct_college} + \beta_8 \cdot \text{bathrooms} + \beta_9 \cdot \text{rooms} + \\ \beta_{10} \cdot \ln(\text{lot_size}) + \beta_{11} \cdot \ln(\text{land_value})\end{aligned}$$

Lastly in model C, we used stepwise selection. The stepwise selection method is a combination of the other two variable selection methods: forward and backwards. Here, the variables can either be added or removed from the selection each time. Model C produced the same final model as model B with the same AIC value.

$$\begin{aligned}\ln(\text{price}) = \beta_0 + \beta_1 \cdot \text{waterfront} + \beta_2 \cdot \text{age} + \beta_3 \cdot \text{new_construct} + \\ \beta_4 \cdot \text{central_air} + \beta_5 \cdot \text{fuel_type} + \beta_6 \cdot \text{living_area} + \\ \beta_7 \cdot \text{pct_college} + \beta_8 \cdot \text{bathrooms} + \beta_9 \cdot \text{rooms} + \\ \beta_{10} \cdot \ln(\text{lot_size}) + \beta_{11} \cdot \ln(\text{land_value})\end{aligned}$$

After obtaining all three models, we performed a 10 fold cross validation to compare the three models' predictive performance (Kuhn and Max, 2008). We were then able to determine the model that provides the best predictive accuracy. From there, we checked the assumptions for the final model we chose, making sure all assumptions were fulfilled.

Results. Three multiple regression models were built using different variable selection approaches to predict the logarithm of housing prices. Their performance was evaluated and compared in terms of predictive accuracy and goodness of fit.

Model	RMSE	Rsquared	AIC	BIC
A	0.2879	0.6015	685.1047	766.9602
B	0.2930	0.5880	684.4653	782.6919
C	0.2922	0.5882	684.4653	782.6919

Fig. 3. Table of results.

Model A (Forward Selection) did the best of the three, with the lowest RMSE (0.2879) and the greatest R^2 (0.6015), which means it was better at making predictions. Its BIC (766.9602) was also the lowest which shows that it strikes a good balance between being easy to understand and being able to explain things. Although it has a slightly higher AIC than the other two models, the difference is not significant.

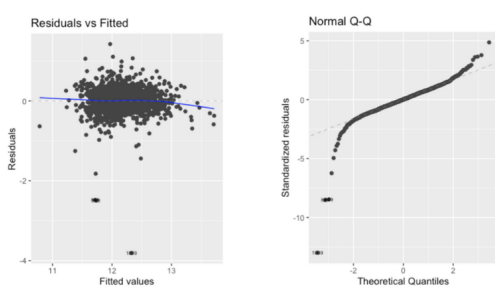

Fig. 4. Diagnostic plots.

Model A's diagnostic plots showed that the residuals were randomly spread out around zero with no clear pattern, indicating the residuals are not systematically related to one another. This means that the assumptions of independence, linearity and homoskedasticity were met. The Q–Q plot indicated that the residuals were nearly normally distributed, with only a few points deviating at the extremes.

The estimated coefficients from Model A showed several important relationships. Houses located on the waterfront were, on average, about 51% more expensive, while central air conditioning was associated with roughly a 5% increase in price. Newly built houses were around 13% cheaper, and each additional year of house age was linked to an estimated 0.1% decline in value. Land value and number of bathrooms both had positive impacts, with increases of approximately 12% and 11% respectively. The number of rooms and lot size also contributed positively to price, though to a smaller extent.

Overall, Model A was selected as the final model since it achieved the best combination of predictive performance and compliance with regression assumptions.

Discussion and Conclusion. Our analysis and models had a few limitations that we could not resolve. First was the possible spatial dependency of our given dataset. The dataset likely had autocorrelations due to geographic proximity, given all the data was gathered in the New York region, which could possibly be skewing our models & results. Another limitation of the dataset was the inability to account for any possibly omitted or 'other' features, as the dataset was not exhaustive of every aspect of the homes. It is possible that other factors, such as neighborhood, proximity to transport or more could strongly influence property value, and is thus a missing component in our analysis.

To help counteract the limitations of our model, several methods could be used to improve our model accuracy given what we were provided. This includes general further exploration into our dataset, and any possible transformations, or identification of any previously unknown relationships. A LASSO regression could be utilised to help improve the accuracy of our models, and avoid any overfitting on specific data points and aspects of the data. Machine learning models could also be implemented, in which underlying relationships could be identified, and more accurate models could be created.

Our analysis found that there were a few key factors that primarily influenced house prices in New York. A household's location, specifically if it is a waterfront property, was found to be a significant indicator of price. House characteristics were also important, as more bathrooms and central air conditioning were great positive indicators of price, and larger living area was also a good indicator for price. An interesting conclusion was that whilst older houses de-value over time, brand-new houses also sold at lower prices, possibly due to construction quality concerns in which newer properties may not have identified building faults yet which come with age. The last important factor was heat type, in which houses with no heating had strong negative correlations with price.

Appendix

References

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